

Graph-Guided Reasoning and Refinement in Diffusion Language Models

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Abstract

Diffusion Language Models (DLMs) offer an alternative to autoregressive language generation by iteratively refining partially masked sequences, enabling revision during generation. However, existing correction mechanisms primarily operate at the token or sequence level and do not explicitly represent dependencies between reasoning steps. We propose a graph-guided reasoning framework that represents intermediate reasoning as a dependency graph and uses this structure to guide denoising and localized error correction. By selectively re-masking only the reasoning subgraph associated with an inconsistency, our approach aims to improve reasoning consistency while reducing unnecessary regeneration and potentially improve computational efficiency. We plan to evaluate the framework on multi-step reasoning benchmarks and compare it against existing diffusion-based reasoning and correction methods.

1 Motivation

Large Language Models (LLMs) have demonstrated strong performance on a wide range of reasoning tasks. However, most LLMs are autoregressive, generating text sequentially from left to right (Li et al., 2026). While sequential generation is simple and effective, once a token has been generated, it cannot be revised. Consequently, errors made in the reasoning process propagate through subsequent steps and negatively impact the final output (Yu et al., 2026). Although techniques such as self-consistency, verification, and post-hoc analysis can mitigate these issues, they often require regenerating large portions of the reasoning process and do not provide a straightforward mechanism for revising specific reasoning steps.

Diffusion Language Models (DLMs) have recently emerged as an alternative in natural language processing (NLP) (Li et al., 2026). Rather than pro-

ducing text token-by-token, DLMs iteratively denoise a partially masked sequence until a complete output is obtained. This iterative process allows for refinement and correction during generation. As a result, DLMs provide a potential framework for error correction and reasoning refinement not possible in autoregressive models (Ye et al., 2024). Recent work has explored diffusion-based reasoning, Chain-of-Thought (CoT) generation, and self-correction mechanisms, demonstrating that refinement during generation can improve reasoning quality (Ye et al., 2024; Huang et al., 2025; Schiff et al., 2026). Several approaches have introduced structure into DLM generation through logical constraints, causal concepts, and graph-based representation (Aman, 2026; Han et al., 2025).

Despite these advances, existing structured diffusion approaches primarily use graphs for reasoning representation, search, or generation guidance (Besta et al., 2024; Han et al., 2025). However, existing correction methods primarily operate through remasking and iterative refinement at the token or sequence level, treating reasoning as unstructured text (Huang et al., 2025; Schiff et al., 2026). As a result, when inconsistencies are detected in the reasoning space, the model lacks an explicit representation of how reasoning steps depend on each other, making it difficult to determine and fix which portions of the reasoning process are responsible for the error and which can be safely preserved. Additionally, correction may require revisiting large regions of otherwise valid reasoning, leading to unnecessary computation and potentially introducing new errors.

We argue that reasoning structure can play an active role in the error-correction process. Many reasoning tasks naturally form dependency graphs in which future conclusions rely on earlier steps while others remain independent. We therefore propose a graph-guided diffusion framework that represents reasoning as a dependency graph. Un-

like previous graph-based reasoning methods that primarily use graphs for reasoning representation, search, or generation control, our approach uses the graph as a mechanism for refinement or correction during diffusion. We hypothesize that graph-guided refinement can improve both efficiency and reliability in diffusion-based reasoning. By using the dependency structure to identify and repair only the minimal reasoning subgraph related to the contradiction, a diffusion language model may avoid unnecessary regeneration while also preserving correct intermediate conclusions. We aim to evaluate whether graph-guided denoising and subgraph-level correction provide improvements in reasoning consistency, computational efficiency, and overall accuracy in DLMs over existing diffusion-based reasoning and correction methods.

In this project, we address the limitations of unstructured token-level correction in DLMs by designing a reasoning graph. Our contributions can be as follows:

- We introduce a novel generation approach that parses intermediate reasoning steps into a directed dependency graph. Unlike existing DLM architectures that denoise in flat sequences, our framework will enable dependency-aware scheduling. This allows independent paths to be refined in parallel while dependent steps wait for their prerequisite steps to stabilize.
- We build a targeted refinement mechanism that uses the reasoning graph to isolate and regenerate inconsistent nodes. By identifying the minimal affected subgraph tied to a detected error, our approach restricts the remasking to only the erroneous nodes and their affected neighbours. This resolves the computational inefficiency of sequence-level correction by preserving mathematically or logically sound reasoning steps and avoiding the introduction of new errors during regeneration.
- We provide a rigorous evaluation to quantify the computational and accuracy metrics of structural refinement. By testing on multi-step reasoning datasets and comparing against token-level self-correction baselines (such as RemeDi and ProSeCo), we show where our framework stands amongst existing ones.

2 Related Works

Reasoning in Diffusion Models Diffusion-of-Thought (Ye et al., 2024) first adapted chain-of-thought reasoning to DLMs, denoising the reasoning trace as a whole rather than emitting it left to right. Every step in the trace stays editable until the final denoising step, so a DLM can revise an earlier incorrect step where an AR decoder cannot. d1 (Zhao et al., 2026) shows that SFT and GRPO-style RL transfer to DLMs and scale reasoning performance further.

Remasking and Error Correction DoT observes that DLMs self-correct across denoising steps as a side effect of iterative refinement. RemeDi (Huang et al., 2025) makes this explicit by predicting a per-token confidence alongside the token distribution, remasking low-confidence tokens and resampling them under richer context, trained with remask-aware SFT followed by RL. ProSeCo (Schiff et al., 2026) instead trains a corrector on the denoiser’s own unmasking errors and interleaves correction steps between unmasking steps. Remask-Don’t-Replace (Yao, 2026) shows that resetting a suspect token to [MASK] and re-denoising beats overwriting it in place, and requires no training to do so. All three detect errors from a local signal, either at token confidence or a per-token corrector score, and correct at token granularity.

Unmasking Scheduling in DLMs Masked diffusion permits the model to unmask tokens in any order. The output tokens, however, carry structural dependencies in most cases. DAPD (Kim et al., 2026) infers token-level dependencies and denoises independent tokens in parallel to maximise throughput. DOS (Zhou et al., 2026) exploits attention matrices to derive inter-token dependencies and emphasises information from unmasked tokens while updating masked ones. DEMASK (Ringel et al., 2026) adds a dependency predictor to estimate conditional influences between masked positions and performs parallel demasking within a dependency bound. These works are focused on token-level structure. Scheduling at the sub-problem level is approached by two works. Planned Diffusion (Israel et al., 2026) first plans using AR decoding, then splits the plan into semantically distinct chunks and denoises them in parallel; the plan encodes independence between chunks but not directed dependency. LogicDiff (Aman, 2026) trains a role prediction head on a frozen LLaDA and unmask in a fixed

logical order. However the template is fixed and not adaptable to specific problems.

Structured Decoding in Autoregressive Models Decomposing generation into structured units is well explored for AR models. Skeleton-of-Thought (Ning et al., 2024) decodes a skeleton of the answer first and then expands each point in parallel, reporting up to $2.39\times$ speedup. It assumes the points are independent, so it degrades on tasks that need sequential reasoning. Plato (Jin et al., 2024) drops that assumption and organises sub-problems into a dependency DAG, decoding a node once its parents are complete. Tree-of-Thoughts (Yao et al., 2023) and Graph-of-Thoughts (Besta et al., 2024) treat reasoning as search over nodes representing partial thoughts and support branching, aggregation and backtracking between them. All of these simulate parallelism on top of a sequential decoder. Additionally an AR model cannot revise a node once it has been processed. A masked DLM gets both these properties natively: independent nodes are unmasked in the same step, and revision needs only a remark.

3 Proposed Methodology

We propose a graph-guided reasoning framework for Diffusion Language Models (DLMs). Given a reasoning task, the model first generates an intermediate reasoning trace, which is represented as a dependency graph. The framework uses this graph both to guide the denoising process and to perform localized error correction.

3.1 Dependency Graph Construction

The intermediate reasoning trace is represented as a dependency graph where nodes correspond to reasoning steps or claims and directed edges represent the dependency relationships between them. Following graph-based reasoning approaches such as Graph of Thoughts (Besta et al., 2024), we represent the reasoning trace as a directed graph ($G = (V, E)$), where an edge $s_i \rightarrow s_j$ indicates that reasoning step s_j requires information established by reasoning step s_i .

The graph is constructed from intermediate reasoning steps generated by the model, with dependencies inferred from explicit references between steps or extracted using a lightweight parsing procedure. Explicit references, shared variables or entities, and semantic dependency classification are used to identify these relationships.

3.2 Dependency-Aware Denoising

The dependency graph is used to investigate dependency-aware denoising schedules. Independent reasoning paths are denoised in parallel, while dependent nodes are refined only after their supporting reasoning nodes have stabilized. This allows the denoising process to take the structure of the reasoning trace into account rather than treating it as an unstructured sequence. Our framework employs an iterative denoising process of diffusion language models while introducing dependency structure to determine the order of refinement.

3.3 Inconsistency Detection

The dependency graph is also used for localized error correction. We introduce an inconsistency detection mechanism that monitors generated reasoning for inconsistencies. For mathematically verifiable reasoning, deterministic verification is used wherever possible to check arithmetic operations, numerical assignments, and derived quantities. For semantic or non-deterministic claims, a Natural Language Inference (NLI)-based verifier is used to identify contradictions between related reasoning steps. NLI determines whether a statement entails, contradicts, or is neutral with respect to another statement (Bowman et al., 2015). In our framework, it is used to identify semantic or logical contradictions between dependent reasoning nodes. The dependency graph restricts this verification to relevant reasoning relationships rather than requiring every pair of reasoning steps to be compared.

3.4 Identifying the Minimal Affected Subgraph

When a contradiction is detected, the framework identifies the smallest dependency subgraph associated with the conflict and selectively remarks only the affected nodes. To identify the subgraph, we draw on the notion of minimal inconsistent subsets, where an inconsistency can be attributed to a minimal subset of conflicting statements (Hunter and Konieczny, 2008). The dependency graph is then used to propagate the conflict to reasoning steps that depend on the inconsistent nodes. The propagation continues while the validity of subsequent nodes depends on the affected reasoning and stops when a node can be independently verified. The resulting set of conflicting and dependent nodes forms the affected reasoning subgraph. This allows

the framework to distinguish between reasoning that must be regenerated and reasoning that can safely be preserved.

3.5 Localized Remasking and Refinement

The resulting subgraph is then re-denoised while preserving the remainder of the reasoning trace. The correction process can be repeated by reconstructing the dependency graph and re-checking for inconsistencies until the reasoning is consistent (or a predefined correction limit is reached). Unlike existing correction methods that often operate at the token or sequence level, this approach performs correction at the level of reasoning structure. Recent diffusion-based correction approaches demonstrate that previously generated tokens can be remasked and refined (Huang et al., 2025; Schiff et al., 2026). Our approach uses the reasoning graph to determine which tokens corresponding to affected reasoning steps should be remasked. We hypothesize that this targeted refinement will be more computationally efficient, as it avoids regenerating large portions of otherwise correct reasoning and focuses computation only on the region responsible for the inconsistency.

4 Datasets

We evaluate our graph-guided reasoning framework and localized error correction on multi-step reasoning benchmarks spanning mathematical and logical domains, enabling both deterministic and semantic verification.

- **GSM8K** (Grade School Math 8K) (Cobbe et al., 2021): A benchmark of 8500 multi-step grade school math word problems. We use GSM8K for deterministic verification, extracting mathematical calculations as reasoning nodes and testing whether localized renoising can correct miscalculations.
- **SVAMP** (Simple Variations on Arithmetic Math Word Problems) (Patel et al., 2021): A benchmark of 1,000 one-unknown arithmetic word problems constructed through controlled variations of problems from ASDiv-A designed to test sensitivity to problem structure. We use SVAMP to evaluate graph-guided error detection and localized remasking under structural variations.
- **ProofWriter** (Tafjord et al., 2021): A logical reasoning benchmark in which natural-

language theories and hypotheses require multi-hop deductive reasoning. Its explicit dependency structures support semantic verification between reasoning nodes, allowing us to test localized remasking for errors in deductive proofs.

5 Evaluation Metrics

To evaluate this framework, we plan to conduct experiments on multi-step reasoning benchmarks, such as GSM8K (Cobbe et al., 2021), and other mathematical and logical reasoning tasks. We will compare our framework against the following baselines and evaluate it using the following metrics:

- **Answer Accuracy:** The proportion of problems for which the model produces the correct final answer.
- **Reasoning Consistency Rate:** The proportion of reasoning traces containing no detected inconsistencies between dependent reasoning steps.
- **Remasking Ratio:** The proportion of reasoning tokens that are remasked during correction.
- **Affected Subgraph Ratio:** The proportion of reasoning nodes identified as affected by an inconsistency.
- **Denoising/Forward Passes:** The number of model denoising or forward passes required during correction.
- **Inference Time:** The wall-clock time required to generate and correct each reasoning trace.

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